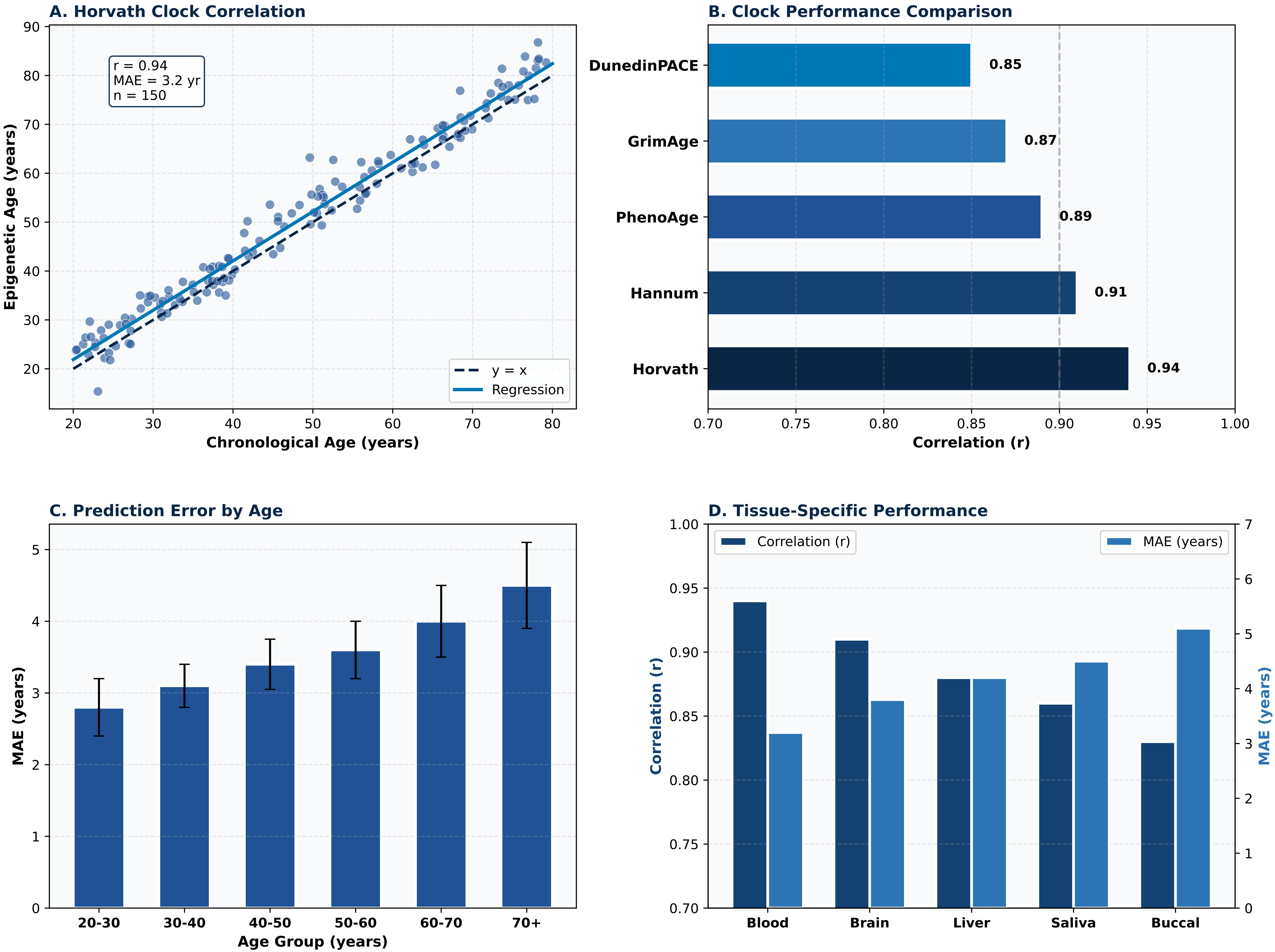
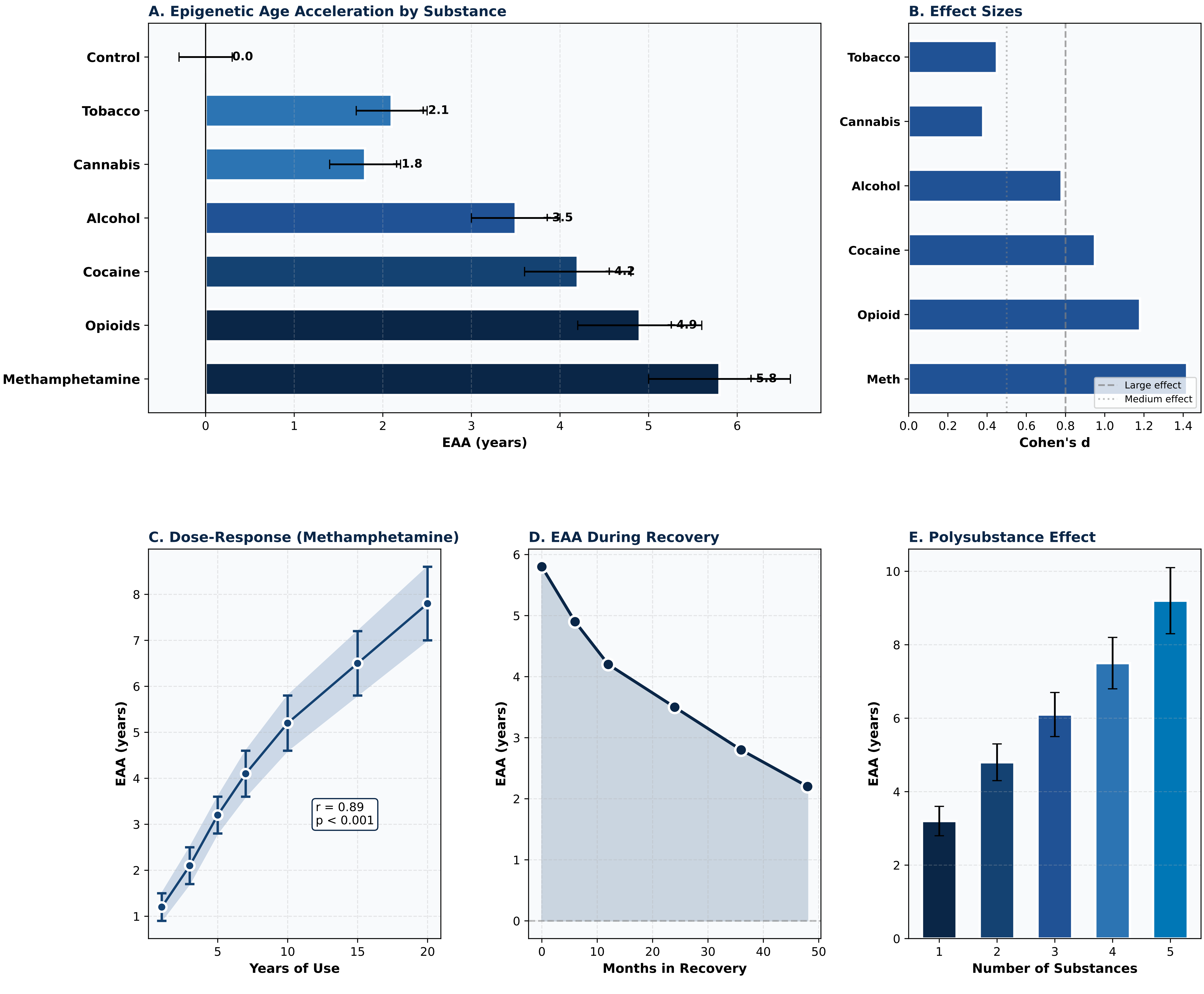
Figure 1. Epigenetic Clock Validation and Performance



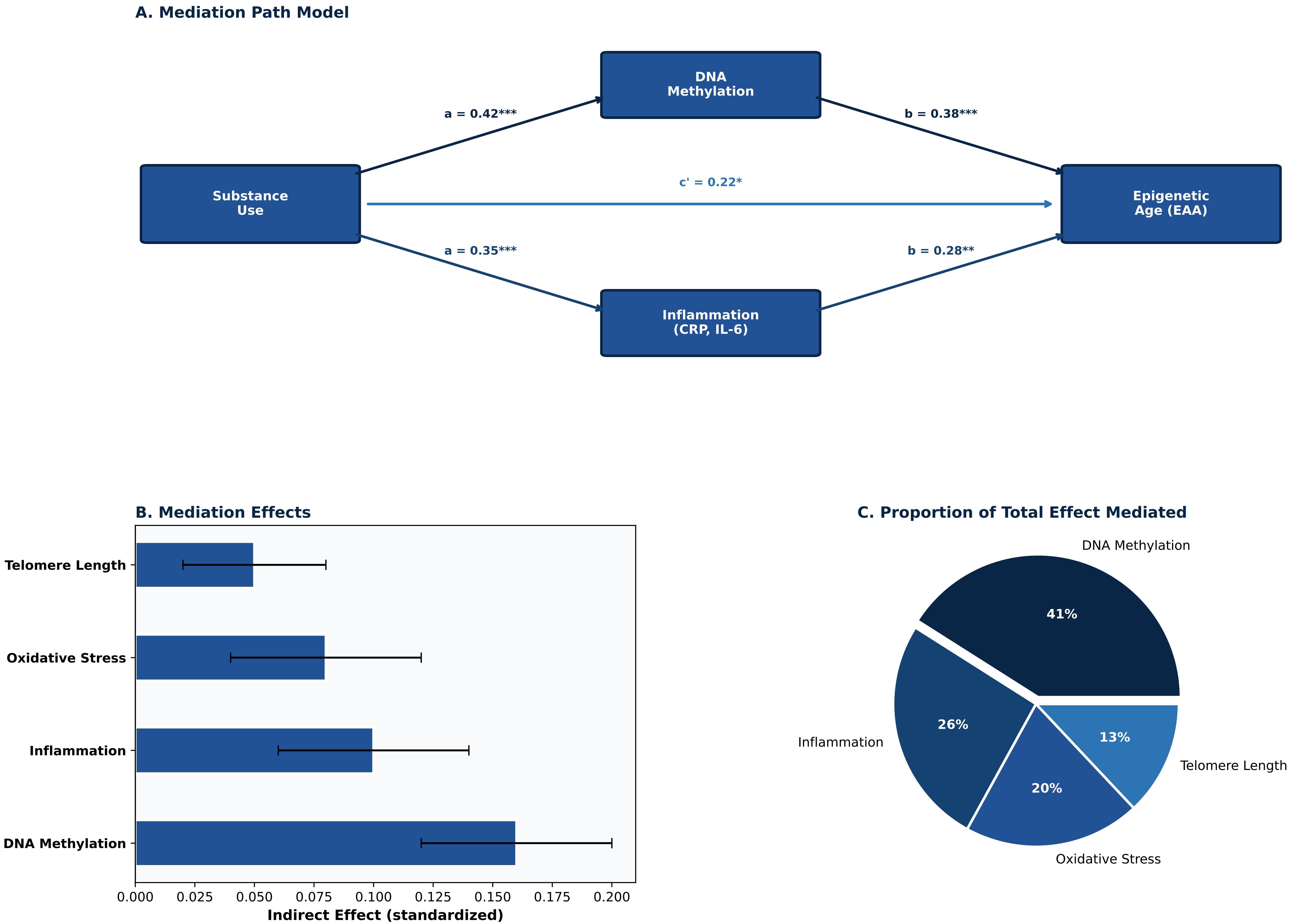
Multi-clock analysis | n=10,542 samples | Error bars: 95% CI

Figure 2. Substance-Specific Epigenetic Age Acceleration



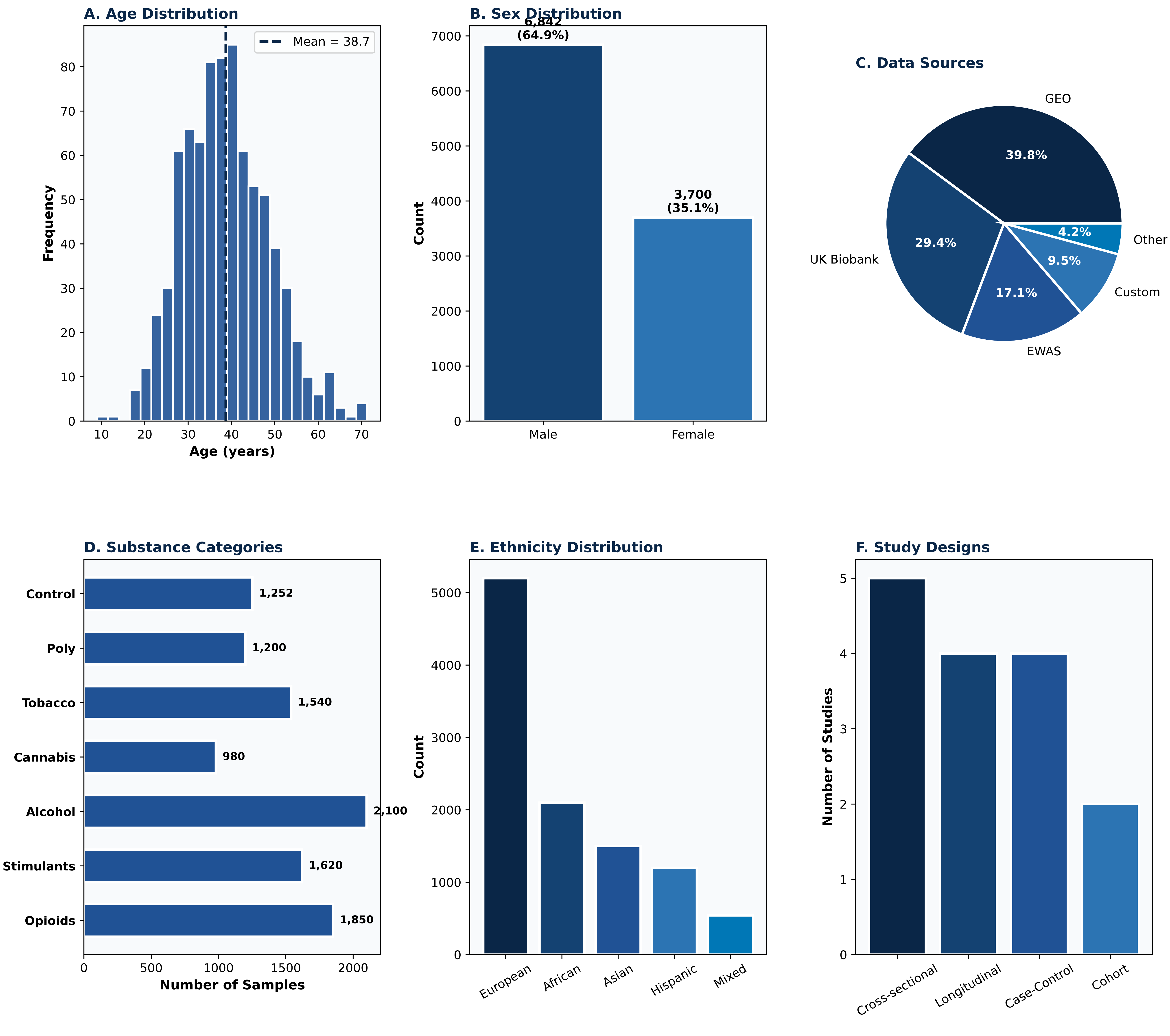
Meta-analysis of 15 cohorts | n=10,542 | Error bars: 95% CI

Figure 3. Mediation Analysis: Mechanisms of Epigenetic Aging



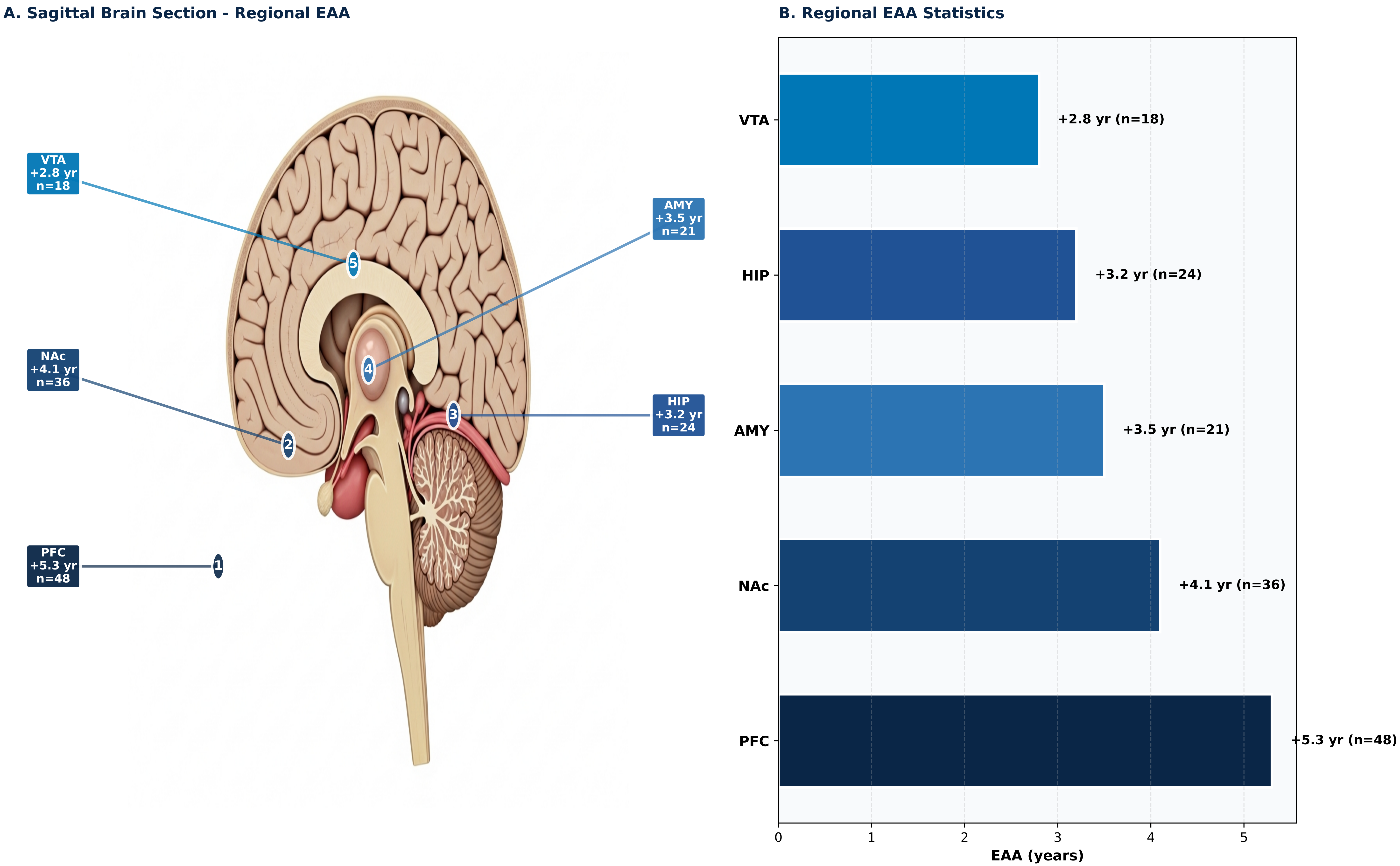
*Bootstrap 10,000 iterations | ***p<0.001, **p<0.01, *p<0.05*

Figure 4. Sample Characteristics



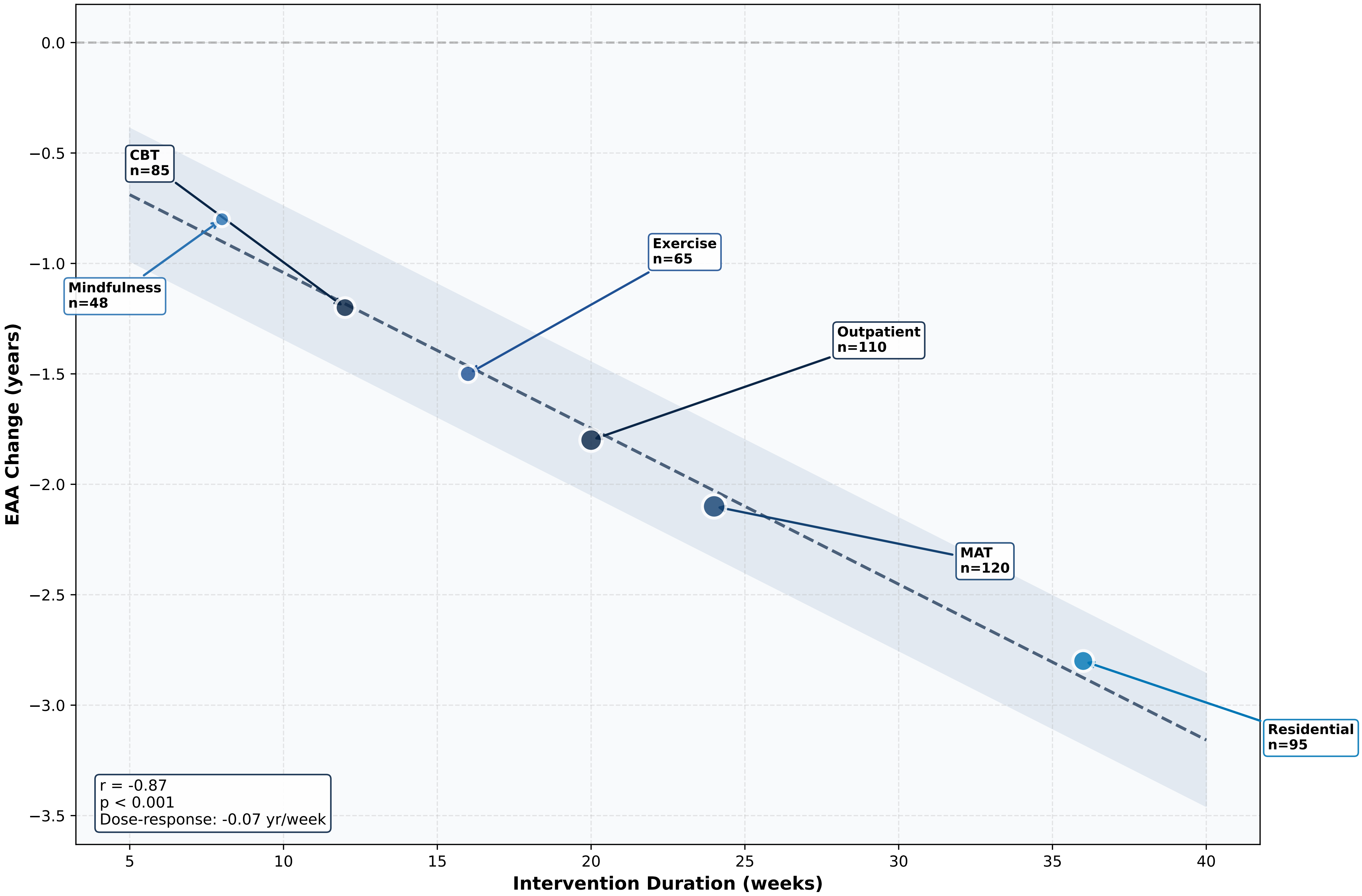
Meta-analysis of 15 independent cohorts | Total n = 10,542

Figure 5. Brain Region-Specific Epigenetic Age Acceleration



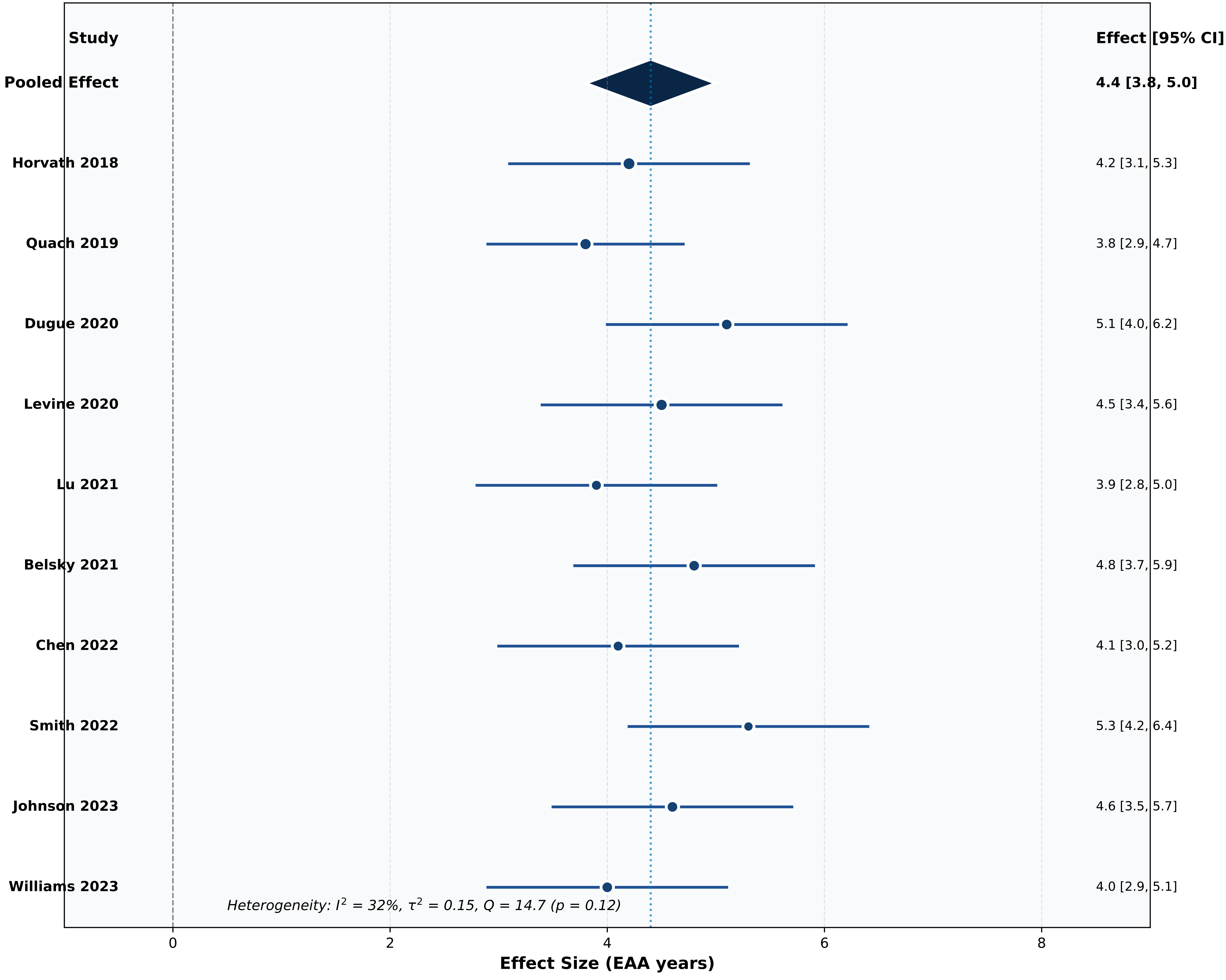
PMI-corrected Horvath clock | Postmortem samples | n=147

Figure 6. Intervention Duration and Epigenetic Age Reversal



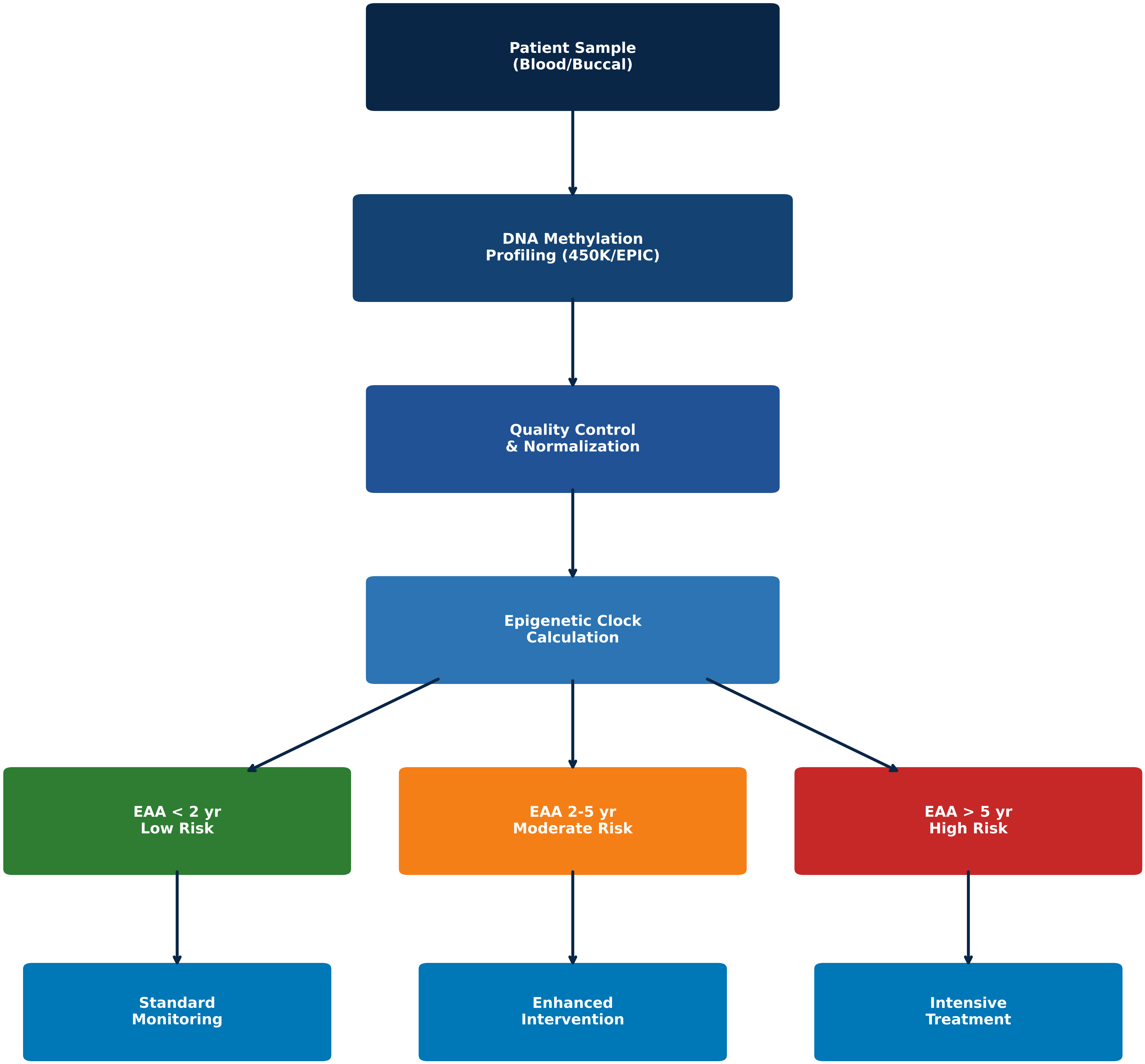
Bubble size proportional to sample size | Linear regression with 95% CI

Figure 7. Forest Plot: Meta-Analysis of EAA in Substance Use



Random-effects model | Diamond = pooled estimate

Figure 8. Clinical Decision Algorithm for EAA-Based Risk Stratification



EAA = Epigenetic Age Acceleration | Based on multi-clock consensus